
Forecasting Chaos Across Model Families: An Out-of-Sample, Reliability-Adjusted Benchmark

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Abstract

We study how different families of models forecast a chaotic mechanical system, and how such a comparison should be scored. We construct a controlled k -pendulum benchmark ($k \in \{1, 2, 3\}$; chaotic for $k \geq 2$) and evaluate four families—large language models (LLMs), a time-series foundation model (Chronos), learned dynamics models (Neural ODE, Hamiltonian and Lagrangian neural networks), and classical integrators (Euler, RK4, symplectic)—on the same out-of-sample held-out trajectories, so that learned models are not scored on their training data. Two design choices shape the results: an out-of-sample protocol, and a reliability-adjusted metric that assigns a fixed error to refusals and parse failures rather than discarding them. Out of sample, the learned dynamics models increase their error by approximately a factor of three relative to in-sample evaluation (~ 0.26 to 0.84 – 1.06 rad) and rank below both a zero-shot time-series model and a zero-shot LLM, and do not clear simple baselines—persistence and a nearest-neighbor analog lookup—built from the same data. Among the black-box methods evaluated, the lowest error is obtained by an LLM (kimi, 0.538 rad), while the known-physics integrators, which are given the exact equations, form a ceiling by construction. The training objective affects out-of-sample error (rollout loss is lower than derivative matching, $p_{\text{Holm}} = 1.6 \times 10^{-7}$), but mixed- k rollout, which is best in-sample, shows no significant out-of-sample effect. A matched, confound-free experiment and a direct inference probe find little evidence that the LLMs infer hidden physical constants; under the conditions of our benchmark they default to Earth-standard values. We report the cost of each family alongside its error: the lowest-error black-box method is also the most expensive per query, whereas the numerical and trained models are free at inference. In-sample evaluation can therefore substantially overestimate learned-dynamics performance, which indicates a need for controlled out-of-sample comparison of forecasting methods.

1 Introduction

The outcome of comparing forecasting methods on a chaotic system depends on several choices: whether models are evaluated in or out of sample, how failures are scored, and what information each model receives. We treat these as benchmark-design questions. Current forecasters span several families—LLMs, time-series foundation models, learned dynamics models, and classical integrators—that are seldom evaluated on one system under identical, out-of-sample conditions. Given the state of a dynamical system at time t , we ask which family most accurately predicts the state at $t + T$, and how that comparison should be scored. We use the k -link pendulum: it is integrable at $k=1$ and chaotic with a positive Lyapunov exponent at $k \geq 2$, its ground truth is computable to high precision, and grading requires no domain expertise. All four families are evaluated on this system under

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identical inputs, horizons, and metrics, with reliability and the structure of chaos treated as part of the evaluation.

Contributions.

1. A **cross-family evaluation protocol** in which all families are scored on the same out-of-sample held-out trajectories, with a reliability-adjusted metric that assigns a fixed error to abstentions rather than discarding them (§3.4, §5.1).
2. An **out-of-sample leaderboard**: tested off their training trajectories the learned models reach 0.84–1.06 rad (from ~ 0.26 in-sample) and do not clear simple baselines (persistence, nearest-neighbor analog lookup), while among the models evaluated the lowest black-box error is obtained by an LLM (kimi, 0.538 rad). Rollout loss reduces error relative to derivative matching, while mixed- k rollout shows no significant out-of-sample effect (§5.4).
3. **Chaos-appropriate analysis**: a predictability-horizon metric and a confound-free system-identification test. An unadjusted regime contrast suggests lower error when constants are hidden; a matched paired experiment and a direct inference probe indicate that this is a difficulty confound rather than parameter inference (§5.6).

2 Related Work

Learned dynamics. Neural ODEs learn a continuous-time vector field from data [Chen et al., 2018]. Hamiltonian Neural Networks impose an energy-conserving inductive bias through a learned Hamiltonian [Greydanus et al., 2019]; Lagrangian Neural Networks enforce the Euler–Lagrange equations through second-order autodifferentiation [Cranmer et al., 2020]. Chaotic systems such as the Lorenz attractor and the double pendulum are standard testbeds [Gilpin, 2021]. We evaluate all three against general-purpose models under matched conditions.

Foundation models as forecasters. LLMs can act as zero-shot time-series forecasters when series are serialized as text [Gruver et al., 2023], and time-series foundation models—Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and Moirai [Woo et al., 2024]—are trained for general forecasting. Zhang and Gilpin [2025] study zero-shot forecasting of chaotic systems with such models. We place these forecasters on the same grid as physics-informed and language models.

LLMs for physical and mathematical reasoning. LLMs perform well on mathematical problem solving [Lewkowycz et al., 2022] and, with chain-of-thought prompting, on multi-step reasoning [Wei et al., 2022]; recent benchmarks probe physics reasoning in LLMs [Xu et al., 2025]. Our task instead requires quantitative trajectory prediction of a chaotic system rather than qualitative or symbolic reasoning.

3 Benchmark Design

Choice of system. We use the k -link pendulum as a controlled system for evaluating methodology rather than as the object of study. It provides a tunable chaos axis (integrable at $k=1$, chaotic at $k \geq 2$) that a single attractor does not; ground truth to high precision; known governing equations, which provide an oracle ceiling and physics-aware baselines; and controllable constants, which allow parameters to be hidden for the system-identification test (§5.6). A less structured system would confound these effects, so the single controlled system is a deliberate design choice.

3.1 Task and systems

Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, the task is to predict $\mathbf{s}_T$ at horizon T . Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t = 10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are produced by RK4 at this step, the RK4 baseline reported later (the same integrator at a $10\times$ coarser $\Delta t = 10^{-2}$) is an oracle ceiling by

Table 1: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
<code>changed_disclosed</code>	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
<code>changed_hidden</code>	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 2: Inputs available to each model family. “Constants” = $(g, L, m, \text{damping})$. Only Chronos receives history; only the learned models receive the true constants in the `hidden` regime.

Family	State $t=0$	History $t<0$	Constants	Equations of motion
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ (all regimes)	— (learned)
Time-series (Chronos)	—	✓ (10 s)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except <code>hidden</code>	—

construction rather than an independent competitor; Euler and the symplectic integrator are distinct known-physics methods. Inputs available to each family are stated in §3.3.

3.2 Regimes and grid

Three regimes test generalization to non-standard constants (Table 1). In `changed_hidden` the constants $(g, L, m, \text{damping})$ are not disclosed to the model, which makes the cell an in-context system-identification problem. An evaluation cell is a tuple (model, k , regime, modality, horizon, prompting); horizons are $\{0.01, 1, 10, 60\}$ s, modalities $\{\text{coords}, \text{image}, \text{image+coords}\}$ (LLMs only), and prompting $\{\text{direct}, \text{chain-of-thought}\}$ (LLMs only). Non-LLM models use the coordinate/direct setting.

Out-of-sample protocol. Learned dynamics models are evaluated off their training data. We generate a shared held-out set (a separate seed from the training trajectories) of 180 initial conditions (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate every family on the identical trajectories, keyed by `movement_id`. The cross-family leaderboard (§5.1) uses horizons $\{1, 10\}$ s, where chaos has begun but error has not saturated (§3.4), giving $180 \times 2 = 360$ cells per model. We report no in-sample numbers and quantify the size of the in-sample effect directly (§5.1).

3.3 Inputs and grouping

Table 2 states what each family receives. Two asymmetries favor the non-LLM models, which makes the comparison conservative for the LLMs: (i) Chronos is the only model that receives the pre- $t=0$ history window; the LLMs and learned models receive only the $t=0$ state. (ii) The learned models receive the true constants in every regime, including `changed_hidden`, because they encode constants as network inputs, whereas the LLMs receive them only when disclosed. Despite these two advantages, the learned models and Chronos rank below an LLM that receives neither (§5.1). A third asymmetry runs in the opposite direction and is larger in magnitude: the LLMs are frontier-scale models with large parameter counts and training budgets, whereas the learned models are small (a 256-wide, 3-layer MLP trained on a single 8 GB GPU; Appendix B). The comparison thus contrasts off-the-shelf frontier models with small learned simulators; the behavior of a substantially larger physics-informed network is outside the scope of this study. We group methods into known-physics (RK4, Euler, symplectic; given the exact equations and constants) and black-box (LLMs, Chronos, learned dynamics). The known-physics methods form a ceiling by construction, and the comparison of interest is among the black-box methods.

3.4 Metrics

The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over links and trajectories; we also track Cartesian bob error, angular-velocity sign match, and energy drift. Three choices are specific to this setting.

Reliability-adjusted error. A cell is answered only if the model returns a parseable, finite prediction. Reporting error over answered cells alone would reward abstention on hard cells. We therefore assign unanswered cells the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform guess); we also report the strict upper bound π . Answering is thus part of the score.

Predictability horizon. For chaotic systems, pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that exceeds τ . The contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. We sweep τ .

System-identification gap. The gap $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$ separates the cost of inferring the constants from being given them. A model that substitutes given numbers is penalized by hiding them ($\Delta > 0$); a model that infers them in context is not ($\Delta \leq 0$). Measuring this requires holding the trajectory fixed and toggling only disclosure; otherwise disclosure is confounded with each regime’s constants and initial conditions. We therefore run a matched experiment (§5.6): 5 non-standard constant sets, 10 trajectories each, $k \in \{1, 2, 3\}$, evaluated under disclosed and hidden conditions on identical trajectories, with a probe that asks the hidden-condition model to report its inferred constants.

4 Methods

LLMs. We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API (Azure AI Foundry deployments; evaluated June 2026), with a structured prompt (system description, disclosed constants where applicable, initial state) and a JSON answer schema (Appendix A). We use a uniform configuration (65,536 max tokens, 300 s timeout) for all three; reasoning traces are read from the dedicated reasoning channel when present. We take a single deterministic sample per cell (temperature 0); no best-of- N aggregation is used, so reliability reflects the raw per-call behavior. The API cost of the out-of-sample LLM evaluation (Parts A and B, 3,960 cells) was \$110.

Time-series foundation models. We run Chronos [Ansari et al., 2024] locally on a single GPU. Chronos is the only model given history: its adapter takes the 10 s pre- $t=0$ window of all $2k$ state channels, resamples it onto a uniform grid of up to 512 points, and forecasts to the target horizon (prediction length ≤ 64 steps). The leaderboard uses `chronos-2`, which forecasts either each channel independently (univariate) or all $2k$ channels jointly (multivariate); this tests, with one model, whether modeling cross-channel coupling helps on a coupled system. For that test we use 20 trajectories per (k, regime) cell and compare the two modes paired on identical trajectories (Wilcoxon signed-rank, bootstrap confidence intervals, Holm correction). The earlier univariate `chronos-t5-large` was used only in a preliminary round; TimesFM is univariate; Moirai’s `uni2ts` could not be installed in our environment.

Learned dynamics. Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with `scipy`. We compare two objectives: derivative matching (MSE on finite-difference $\dot{s}$) and rollout loss (MSE on the state after k Euler steps, with gradients through the unroll). We also use mixed- k rollout, which trains on several window lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$). The LNN’s loss requires second-order autodifferentiation, which Apple’s MPS backend does not run; we train it at full scale (67k samples, hidden 256) on a CUDA GPU. All learned models are trained on the same nine trajectories—one per (k, regime) cell (Appendix B)—a deliberately small training budget whose effect we return to in the Limitations.

Integrators. Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations of motion; their error is from discretization alone. RK4 is the data-generating process and serves as an oracle ceiling rather than an independent competitor.

Baselines. We add four simple baselines as reference floors, evaluated on the same grid; persistence and constant-velocity are trivial, while the linearized and nearest-neighbor baselines are stronger

references (nearest-neighbor is a non-parametric analog forecast, a method used operationally in weather prediction). *Persistence* predicts $\theta_T = \theta_0$; *constant-velocity* predicts $\theta_T = \theta_0 + \omega_0 T$; *linearized* integrates the small-angle linearization about the downward equilibrium (matrix exponential of the numerical Jacobian of the dynamics, using the cell’s true constants); and *nearest-neighbor* returns the state reached T later by the closest state (angle-wrapped distance) in the matching (k , regime) training trajectory—a lookup on the same data the learned models were trained on.

5 Results

5.1 Out-of-sample leaderboard

Table 3 reports the out-of-sample leaderboard on the shared held-out set (360 cells: 180 trajectories $\times$ horizons $\{1, 10\}$ s), using the reliability-adjusted metric so that an unanswered or missing cell is scored at $\pi/2$. Every model is scored on the same grid; the Answered column gives the fraction of cells for which a model returned a valid prediction (only the LNN, whose autograd-ODE integration diverges to NaN on 55% of cells, falls below 100%). LLMs are scored on their direct-prompt forecasts. Confidence intervals are 95% paired bootstrap over trajectories ($B=10^4$); for the LLMs they reflect variation across trajectories only, not per-call sampling, since we take a single temperature-0 sample per cell and hosted models are not perfectly deterministic.

The classical integrators have the lowest error: RK4 (the data-generating process, hence an oracle ceiling) at 0.059, symplectic at 0.205, Euler at 0.405. Among the black-box methods evaluated, the lowest error is obtained by *kimi* (0.538 rad, 99% answered), below both Chronos modes and all learned dynamics models. The four baselines locate the floor. *kimi* is below every baseline with a non-overlapping confidence interval (the closest, nearest-neighbor, is 0.725), so its standing is not an artifact of the scoring; both Chronos modes are also below the best baseline in point estimate. The learned dynamics models are not: in point estimates the nearest-neighbor analog lookup (0.725) is below *every* learned model, the linearized-pendulum baseline (0.855) is below all but the best, and persistence (1.004) is below the plain Neural ODE, HNN, and LNN, with intervals overlapping only for the closest comparisons. Out of sample the learned models therefore do not clear these simple baselines—in several cases not even persistence—despite the baselines using the same training data. They score ~ 0.26 rad under in-sample evaluation and reach 0.84–1.06 rad here (the lowest, mixed- k rollout Neural ODE, ranks 7th); this factor of approximately three indicates that their in-sample strength on this benchmark is largely attributable to evaluation on their training trajectories.

Effect of the reliability adjustment. *grok* answers 82% of the grid and *lln* 45%; scoring those abstentions at $\pi/2$, rather than averaging over answered cells, places them at ranks 8 and 14. Averaging over answered cells alone would rank the least reliable models higher.

5.2 Behavior by horizon

Out-of-sample error by horizon appears in Appendix D (Table 9). At 1 s the ordering follows the leaderboard: the integrators are near-exact, *kimi* (0.457) has the lowest black-box error, and the learned models are at 0.76–0.97. At 10 s all models are in the chaotic regime, and the ordering is stable: *kimi* (0.619) remains lowest among black-box methods while the learned models reach 0.91–1.26. The mixed- k Neural ODE, which is the in-sample standout at 1–10 s, is mid-pack here (0.779 / 0.908). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the leaderboard is the range over which the families remain separable.

5.3 Breakdown by k and by regime

Tables 7 and 8 (Appendix C) report out-of-sample error by number of links and by regime. Error increases monotonically with k for every family (for *kimi*, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The changed_hidden regime has the lowest error for almost every model (for *kimi*, 0.46 hidden vs. 0.64 normal; the learned models are lower by 0.5–0.8 rad relative to normal); its low-gravity, short-rod constants produce slower, less-divergent motion over a fixed horizon, so the unadjusted

Table 3: Out-of-sample leaderboard on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at $\pi/2$; 95% bootstrap CI. RK4 is the data-generating process, hence an oracle ceiling rather than a competitor. Methods new in this work are marked $\dagger$; simple baselines (floors) are marked $\ddagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (oracle)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor $\ddagger$	0.725	[0.657, 0.797]	100%	baseline
7	neural-ode-rollout-mixed $\dagger$	0.844	[0.770, 0.920]	100%	learned
8	linearized $\ddagger$	0.855	[0.774, 0.939]	100%	baseline
9	neural-ode-rollout	0.870	[0.795, 0.947]	100%	learned
10	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
11	hnn-rollout	0.922	[0.847, 0.998]	100%	learned
12	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
13	hnn-rollout-mixed $\dagger$	1.003	[0.923, 1.084]	100%	learned
14	persistence $\ddagger$	1.004	[0.934, 1.078]	100%	baseline
15	neural-ode	1.028	[0.953, 1.107]	100%	learned
16	hnn	1.064	[0.982, 1.145]	100%	learned
17	constant-velocity $\ddagger$	1.319	[1.241, 1.396]	100%	baseline
18	lnn $\dagger$ (full-scale)	1.421	[1.363, 1.481]	45%	learned

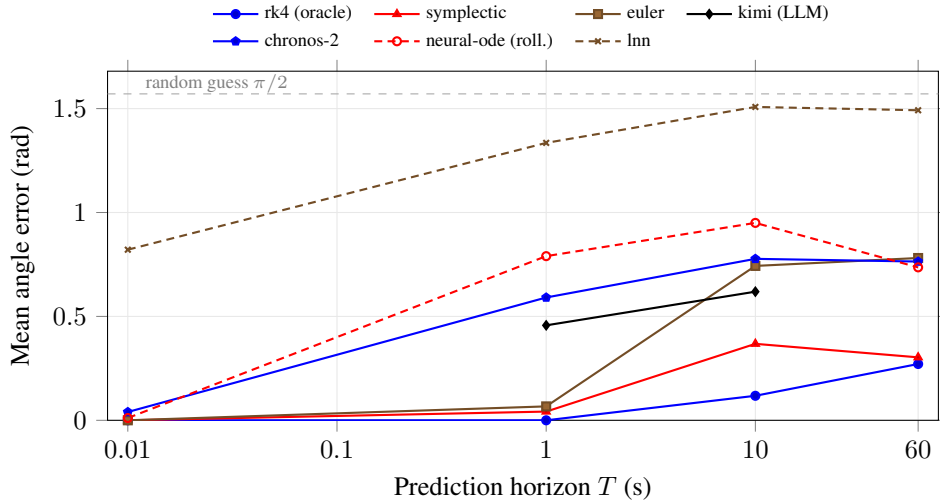


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs were evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 4: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.028	—
	rollout (single- k)	0.870	sig. ($p_{\text{Holm}}=1.6 \times 10^{-7}$)
	mixed- k	0.844	n.s. ($p_{\text{Holm}}=0.089$)
HNN	derivative MSE	1.064	—
	rollout (single- k)	0.922	(rollout lower)
	mixed- k	1.003	n.s. ($p_{\text{Holm}}=0.088$)

“hidden helps” contrast is a difficulty confound (§5.6). Re-estimating the leaderboard confidence intervals with a cluster bootstrap over whole held-out trajectories changes no interval by more than ~ 0.03 rad and changes no ranking, so the gaps are not an artifact of treating correlated cells as independent.

5.4 Training objective

Table 4 reports the effect of the loss with architecture fixed. Rollout loss has lower error than derivative matching: for the Neural ODE, 1.028 drops to 0.870 under a single- k rollout loss, and in a powered paired study ($n=720$ trajectories, Wilcoxon signed-rank) this difference is significant ($p_{\text{Holm}}=1.6 \times 10^{-7}$). Penalizing trajectory error through an unrolled integrator, rather than instantaneous derivative error, optimizes the quantity the benchmark measures. The in-sample advantage of mixed- k rollout does not hold out of sample: for the Neural ODE it is indistinguishable from single- k (0.844 vs. 0.870; $p_{\text{Holm}}=0.089$), and for the HNN it is higher (1.003 vs. 0.922; $p_{\text{Holm}}=0.088$). Full-scale CUDA training of the LNN gives the highest error (1.421), with 55% of cells diverging to NaN under autograd-ODE integration. We do not claim that LNNs are intrinsically weak; under our implementation and protocol the LNN was numerically unstable and generalized poorly out of sample.

5.5 Physical fidelity: energy conservation and stability

Pointwise angle error is not the axis the Hamiltonian and Lagrangian networks are designed for; those architectures target structure preservation. We therefore evaluate energy conservation directly on the undamped (normal) regime, where the true energy is constant and HNN/LNN are well specified. Table 5 reports the mean absolute energy drift over a 10 s rollout, normalized by the fixed scale $E^*=g \sum_i m_i L_i$ (0 = perfect conservation), and the fraction of rollouts that stay bounded.

Three points follow. First, the inductive bias does not translate into physical fidelity: the symplectic integrator (a classical structure-preserving method) drifts by 0.066, while the lowest-drift learned model is 0.78—worse than a plain Euler integrator (0.494) and an order of magnitude above symplectic. Second, this is a property of the learned energy function, not the rollout integrator: rolling the HNN out with a *symplectic* integrator leaves its drift essentially unchanged (e.g. 2.04 vs. 2.07 for hnn-rollout; 0.79 vs. 0.87 for hnn-rollout-mixed), because it conserves its *learned* Hamiltonian exactly but that Hamiltonian is a poor match to the true energy. Third, the Hamiltonian bias gives no advantage over a plain Neural ODE at matched objective (neural-ode-rollout-mixed 0.78 $\approx$ hnn-rollout-mixed 0.87); what reduces drift is the rollout objective, not the structure. The LNN additionally fails on stability, with only 27% of rollouts bounded. Even on the axis they are built for, the learned models do not match the classical integrators. (As for the leaderboard, the learned energy functions are estimated from nine trajectories; §7.)

5.6 Chaos-appropriate analysis

Predictability horizon. Appendix D (Table 10) reports the longest horizon each model tracks contiguously below a threshold τ , over the ladder $\{0.01, 1, 10, 60\}$ s available for the non-LLM models. Only the numerical integrators stay below strict thresholds past 1 s: RK4 tracks to 10 s at $\tau=0.25$ and 60 s at $\tau=0.5$, symplectic to 60 s at $\tau=0.5$, Euler to 60 s only at $\tau=1.0$. Every learned

Table 5: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^*=g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. Rolling the HNN out with a symplectic integrator gives near-identical drift, so the error is in the learned Hamiltonian, not the integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

model, including the rollout and mixed- k Neural ODEs, returns to the first rung (0.01 s) at all strict thresholds and reaches 60 s only at $\tau=1.0$ ($\sim 57^\circ$), where the threshold is uninformative. The LLMs were evaluated only at $\{1, 10\}$ s, so their ladder is coarser; within it, only kimi is below 0.5 rad at 1 s. Within our thresholds, no data-driven model exceeds the predictability horizon of the numerical integrators.

Unadjusted regime contrast. On the shared held-out set the three regimes differ in both their constants and their initial conditions, so a regime contrast conflates disclosure with difficulty. The unadjusted contrast is large: pooled across LLMs, the hidden regime has lower error than the disclosed regime (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524 , $p_{\text{Holm}} < 10^{-3}$). This is the “LLMs do better when constants are hidden” pattern reported in earlier rounds.

Matched experiment. We ran the confound-free test of §3.4: 5 non-standard constant sets, 10 trajectories each, $k \in \{1, 2, 3\}$, scored under disclosed and hidden conditions on identical trajectories, paired by trajectory (Table 6). The effect reverses. Hiding the constants increases error ($\Delta_{\text{sysid}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$); at $k=1$ disclosure helps ($+0.297$, $p_{\text{Holm}} < 10^{-3}$) and at the 1 s horizon as well ($+0.145$, $p_{\text{Holm}} < 10^{-3}$), with a small reverse effect at $k=3$ (-0.055 , $p_{\text{Holm}} = 0.016$). Per model, kimi is penalized by hiding ($\Delta = +0.211$), grok mildly ($+0.038$), and deepseek is flat (-0.042). The advantage of the hidden regime in the unadjusted contrast is therefore a difficulty confound: the hidden-regime trajectories are easier.

Inference probe. In the hidden condition we also asked each model to report its inferred constants. The models do not recover them: against true $g \in [1.6, 12]$ the inferred- g MAE is 1.81 (deepseek), 1.88 (grok), and 2.12 (kimi). The pattern is consistent across models: they default to Earth-standard values (deepseek reports $g \approx 9.81$ on nearly every cell; grok and kimi report $L=m=1.0$ on nearly every cell, a constant output for which the rank correlation with truth is undefined). The one nonzero signal is deepseek’s rod lengths, which correlate with truth at Spearman 0.11 ($p=0.007$). We therefore find little evidence that the accuracy difference between regimes is caused by in-context system identification.

5.7 Univariate versus multivariate time-series modeling

The leaderboard uses one trajectory per cell, which resolves the large gaps between families but not small effects. To test whether a time-series model that shares information across the $2k$ state channels benefits from their coupling, we compare Chronos-2 in univariate and multivariate modes with 20 trajectories per (k, regime) cell ($n=720$ paired samples) on identical trajectories. The two modes are not distinguishable: mean error 0.549 (uni, 95% CI $[0.505, 0.593]$) vs. 0.543 (multi, $[0.498, 0.589]$). The paired difference is $+0.005$ rad overall, with no significant effect at any k or at horizons 1 s and 10 s; the per- k trend at $n=1$ (multivariate appearing to help most at $k=3$) goes to zero at $n=720$. A nominal overall $p=0.025$ is driven by a sub-0.01 rad bias at the 0.01 s horizon and does not survive

Table 6: Matched, confound-free system-identification gap (rad), paired on identical trajectories. $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected.

Model	disclosed	hidden	Δ_{sysid}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

Holm correction. We report a null result: cross-channel coupling gives no measurable benefit for time-series forecasting of this system.

6 Discussion

In-sample evaluation. Scored on held-out rather than training trajectories, the learned dynamics models triple their error and fall below a zero-shot time-series model, a zero-shot LLM, and simple baselines. The “small fitted model beats general-purpose forecasters” conclusion an in-sample leaderboard invites does not survive; a learned simulator must be reported on trajectories it never saw. What transfers among the learned models is the rollout objective, not the architecture or the mixed- k recipe.

Reliability. Scoring abstentions at $\pi/2$, rather than averaging over answered cells, is what places grok and the LNN low; chain-of-thought does not help (paired CoT-vs-direct differences are insignificant). A fixed budget of reasoning tokens does not substitute for sequential integration.

System identification. The apparent “hidden helps” effect is a difficulty confound: on matched trajectories, hiding the constants slightly increases error, and the probe shows the models default to Earth-standard values rather than inferring the true ones. We therefore find little evidence of in-context system identification, and caution against reading regime contrasts as inference without a matched control.

Cost and scope. The LLM evaluation is the only family billed per query (\$110, $\sim$ \$0.03/forecast); the others are free at inference, so the lowest-error black-box method is also the most expensive. The quantitative ranking is specific to this system, but the protocol is not—in-sample optimism, reliability-adjusted scoring, and the matched disclosure control are system-agnostic, and the harness is substrate-parametrized—so applying it to other chaotic systems is a direct extension, one we expect to change the ranking but not the methodology.

7 Limitations

All results use a single held-out seed; additional seeds would tighten the error bars. Paired tests are Holm-corrected within each family (objective, multivariate, system-ID) but not across the paper, so marginal results—for example the overall system-identification gap ($p_{\text{Holm}}=0.034$)—should be read as suggestive. Predictions are single-endpoint; a full-rollout protocol would let the predictability horizon be measured directly. The metric is pointwise; we report energy conservation (\$5.5) but leave fuller probabilistic (CRPS/ensemble) and attractor-distribution scoring, and a noisy-initial-condition version, to future work. The comparison is structurally asymmetric: the learned models are trained on the k -pendulum family and tested on held-out trajectories of it, whereas the LLMs and Chronos are zero-shot. The learned models are trained on only nine trajectories (one per (k , regime) cell), a small budget for a chaotic system; the finding that they do not exceed simple baselines is specific to this training budget, and a data- or capacity-scaling study is needed to separate architecture from data limitations. The Hamiltonian and Lagrangian architectures also conserve energy by construction and

cannot represent dissipation, so the two damped regimes disadvantage them in the pooled leaderboard independently of quality; the conservation test (§5.5) is therefore run only on the undamped regime. The pendulum equations are common in training corpora, so contamination cannot be excluded; the hidden-constants regime mitigates this only in part. Finally, the numerical results are demonstrated on a single system and three LLMs; extending them to other chaotic systems (Lorenz, driven pendulum) and a broader model set (e.g. Qwen-VL, TimesFM, Moirai) through the same harness is left to future work.

8 Conclusion

On a controlled k -pendulum benchmark evaluated on shared out-of-sample trajectories, in-sample evaluation of learned dynamics is optimistic: off their training data the learned models triple their error and do not exceed simple baselines, so the lowest black-box error is an LLM (kimi) while the numerical integrators remain the ceiling. The training objective, not the architecture, transfers; reliability-adjusted scoring and a matched, confound-free control overturn the naive reading and find little evidence of in-context system identification. The durable contribution is the protocol, not the version-specific ranking.

Reproducibility. Code, prompts (Appendix A), trained weights, the shared held-out specification, and per-cell checkpoints are released with the benchmark. Training trajectories use seed 42; the held-out evaluation set uses a disjoint seed (20260620), 180 trajectories (60 per regime). LLMs (kimi-k2.6, grok-4-1-fast-reasoning, deepseek-v4-pro, via Azure AI Foundry, evaluated June 2026) are sampled once at temperature 0; hosted deployments may change over time and temperature-0 API sampling is not guaranteed to be reproducible, so the exact LLM numbers may not be reproducible from the API alone, although the protocol and all local (non-LLM) results are. A cell is answered if it returns a parseable, finite prediction; otherwise it is assigned $\pi/2$. All local training and evaluation run on a single consumer GPU (8 GB); no proprietary training data is used.

Author Contributions

S.N.K. (lead author, corresponding) designed the benchmark, led the evaluation harness and the learned-dynamics, time-series, and analysis pipelines, ran the local out-of-sample and significance experiments, and wrote the manuscript. T.S. helped plan and build the benchmark harness. S.S. ran the large language model (API) evaluations. Author order reflects relative contribution. All authors reviewed and approved the final manuscript.

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A Prompt templates

The prompts (from `bench/prompts.py`) are reproduced below.

System prompt (direct / no-CoT).

You are a careful physicist predicting the future state of a k -pendulum (a chain of k point masses on massless rods, hinged in a planar chain, swinging under gravity with viscous damping). Respond with ONLY a single JSON object on one line, no prose, no code fences. Schema: `{"theta": [...k floats..], "omega": [...k floats..]}` where `theta_i` is in radians (link i measured from straight-down, positive counter-clockwise) and `omega_i` is angular velocity in rad/s.

System prompt (chain-of-thought). Same preamble, then: “Reason step-by-step about the dynamics, then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a single JSON object: `{"theta": [...k floats..], "omega": [...k floats..]}`.”

User prompt (coordinate modality).

```
Number of links  $k = \{k\}$ 
Physical constants:
   $g$  (gravity,  $\text{m/s}^2$ ):  $\{g\}$ 
   $L$  (rod lengths,  $\text{m}$ ):  $\{L\}$ 
   $m$  (point masses,  $\text{kg}$ ):  $\{m\}$ 
  damping (viscous coefficient on  $\omega$ ):  $\{\text{damping}\}$ 
Initial state at  $t=0$ :
   $\theta_0$  (rad):  $\{\theta_0\}$ 
   $\omega_0$  (rad/s):  $\{\omega_0\}$ 
Predict the state at  $t = \{T\}$  seconds.
Return  $\theta$  and  $\omega$  as length- $\{k\}$  JSON arrays.
```

Hidden-constants regime. The constants block above is replaced by:

Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point masses, and damping are NOT disclosed. You must infer dynamics from the initial conditions alone.

Parsing and scoring. Each response is stripped of code fences and (for CoT) of everything outside the `<answer>...</answer>` tags; the first `{...}` object is JSON-parsed and validated to contain length- k θ and ω arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

Table 7: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (oracle)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
neural-ode-rollout-mixed	0.616	0.868	1.047
neural-ode-rollout	0.661	0.834	1.116
grok-4-1-fast-reasoning	0.366	1.030	1.247
hnn-rollout	0.508	0.921	1.336
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout-mixed	0.516	1.083	1.410
neural-ode	0.509	1.167	1.409
hnn	0.491	1.230	1.470
lnn	1.146	1.547	1.571

Table 8: Out-of-sample error by regime. `changed_hidden` has the lowest error for almost every model; its low-gravity/short-rod constants give slower, less-divergent motion, which is why the unadjusted regime contrast (§5.6) is a difficulty confound rather than evidence of in-context system identification.

Model	normal	disclosed	hidden
rk4 (oracle)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
neural-ode-rollout-mixed	1.260	0.836	0.434
neural-ode-rollout	1.281	0.894	0.436
grok-4-1-fast-reasoning	1.096	0.959	0.588
hnn-rollout	1.087	1.115	0.564
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout-mixed	1.244	1.172	0.592
neural-ode	1.221	1.095	0.770
hnn	1.262	1.303	0.626
lnn	1.511	1.408	1.345

B Training and compute

All learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate, on a single NVIDIA GTX 1070 Ti (8 GB). Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at $\Delta t = 0.01$ s from the 10^{-3} s ground truth. These samples are drawn from *nine* trajectories—one per (k , regime) cell—so, while the sample count is large, the number of distinct trajectories the learned models see is small.

C Per- k and per-regime breakdowns

Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons $\{1, 10\}$ s), by number of links (Table 7) and by regime (Table 8); models in leaderboard order. Cells a model never produced (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$, so each row is a mean over the same grid as Table 3.

D Additional result tables

Full per-model values for the horizon breakdown (§5.2) and the predictability horizon (§5.6); the key numbers are stated inline in the body.

Table 9: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (oracle)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimik2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
neural-ode-rollout-mixed	0.779	0.908
neural-ode-rollout	0.790	0.950
hnn-rollout	0.758	1.085
grok-4-1-fast-reasoning	0.905	0.857
neural-ode	0.793	1.264
deepseek-v4-pro	0.996	0.909
lnn (full-scale)	1.335	1.508

Table 10: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better.

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (oracle)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	60
neural-ode-rollout-mixed	0.01	0.01	0.01	60
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	1.0
lnn	0.01	0.01	0.01	0.01